AP CALCULUS AB

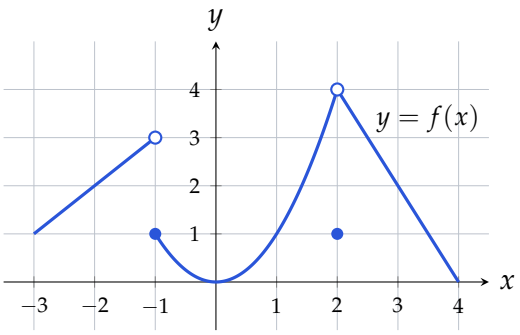
Unit 1 • Lesson 1.3 practice

Lesson 1.3 • Estimating Limit Values from Graphs

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.



The graph of f: a line segment for $-3 \leq x < -1$, $y = x^2$ for $-1 \leq x < 2$, a line segment for $2 < x \leq 4$; $f(-1) = 1$, $f(2) = 1$.

Question 1

NO CALCULATOR

Using the graph of f, what is $\lim_{x \to -1^-} f(x)$?

(A) 3
(B) 1
(C) 2
(D) It does not exist.

Question 2

NO CALCULATOR

Using the graph of f, what is $\lim_{x \to 2} f(x)$?

(A) 1
(B) 2
(C) 4
(D) It does not exist.

Question 3

NO CALCULATOR

Using the graph of f, what is $f(2)$?

(A) 4
(B) 1
(C) 0
(D) undefined

Question 4 NO CALCULATOR

For which value of x in $(-3, 4)$ does $\lim f(x)$ fail to exist?

(A) $x = 2$
(B) $x = 0$
(C) $x = 3$
(D) $x = -1$

Question 5 (graph) NO CALCULATOR

Use the graph of f above. What is $\lim_{x \to -3^+} f(x)$?

(A) 1
(B) -3
(C) 0
(D) It does not exist.

Question 6 (Free response) NO CALCULATOR

Use the graph of f above.

(a) Find $\lim_{x \to 3} f(x)$ and $\lim_{x \to 0} f(x)$.
(b) Find $\lim_{x \to 1} \big(f(x) + 2x\big)$.
(c) Find $\lim_{x \to 2} x\, f(x)$.